

Survival Strategy Framework

Synthetic Retention Demo

Validation Summary (Board-Ready)

Author	Andrew R. Goad
Document Type	Independent Validation and Acceptance Artifact
Validation Scope	Synthetic time-to-event framework, CoxPH modeling, segmentation, scenario simulation, controls, and evidence
Run ID	20260722T171148Z_e580c21e
Run Date	July 22, 2026
Primary Purpose	Determine fitness for controlled portfolio demonstration and synthetic sensitivity analysis
----- EXECUTIVE SNAPSHOT -----	
What Was Validated	A deterministic Python framework spanning survival input governance, K-Means personas, regularized CoxPH modeling, risk targeting, dependency-safe same-cohort scenarios, calibration, PH review, and automated evidence.
Validation Posture	ACCEPTED WITH DOCUMENTED REVIEW for controlled synthetic strategy simulation and public portfolio demonstration.
Key Evidence	7,500 records; 2,964 events; 5-fold CV concordance 0.827; 1,875-record target cohort; six control/improvement/stress scenarios; 42 archived artifacts.
Primary Review Item	auto_renew_flag produced a raw PH p-value of 0.0278; Holm and BH adjusted p-values were 0.333, and a stratified sensitivity model preserved scenario ranking and 93.7% of target-cohort membership.
Reader Takeaway	The system is materially more than a CoxPH model: it is a governed, reproducible, scenario-capable analytical framework with documented assumptions, controls, evidence, and explicit non-production boundaries.
This validation confirms fitness for the documented synthetic demonstration purpose. It does not approve production retention decisions, causal treatment claims, financial forecasts, or deployment without domain calibration and model-risk governance.	

1. EXECUTIVE SUMMARY

The Survival Strategy Framework has been validated as a deterministic, configurable, and evidence-producing time-to-event decision framework. The accepted implementation separates descriptive persona discovery from Cox proportional hazards modeling, uses train-only standardization within cross-validation, selects a governed top-quartile risk cohort, applies control, favorable, and adverse scenarios to the same record identifiers, reconstructs engineered feature dependencies before re-scoring, and archives both executive and technical evidence.

The final run produced a PASS_WITH_REVIEW conclusion rather than an unrestricted pass. This is the appropriate posture: all structural, computational, reproducibility, discrimination, calibration, scenario-direction, and reporting controls passed, while one raw proportional-hazards diagnostic remained documented for review. Multiplicity-adjusted tests passed, and a stratified sensitivity analysis showed that the review item did not materially alter the decision narrative.

Validation Dimension	Final Evidence	Conclusion
Execution integrity	Seven internal self-tests and 18 run-level acceptance checks completed; no failed checks.	Satisfied
Reproducibility	Persona assignments, coefficients, CV, calibration, PH evidence, target cohort, scenarios, and curves reproduced exactly.	Satisfied
Discrimination	5-fold mean concordance 0.827 (SD 0.007; range 0.820-0.836).	Strong for synthetic demonstration
Calibration	Overall errors at 6, 12, and 18 months were 0.95, 1.23, and 1.39 percentage points; maximum risk-group error was 7.08%.	Satisfied
PH assumptions	One raw review item; adjusted p-values pass; stratified sensitivity preserved ranking.	Accepted with review
Persona evidence	Silhouette 0.161; mean stability ARI 0.992; descriptive segments show distinct lifecycle profiles.	Satisfied within descriptive purpose
Scenario governance	Same 1,875 IDs; no-change control neutral; improvement scenarios favorable; stress scenario adverse.	Satisfied
Evidence and lineage	42 artifacts including configuration, model evidence, calibration, scenarios, archives, PPTX, and PDF.	Satisfied

FINAL VALIDATION POSTURE: Accepted with documented review for controlled synthetic time-to-event strategy simulation, educational use, and public portfolio demonstration.

2. TABLE OF CONTENTS / DOCUMENT NAVIGATION

Section	Title	Purpose
1	Executive Summary	States the final validation posture and the basis for acceptance.
2	Document Navigation	Provides the section map.
3	Validation Approach	Defines validation philosophy, methodology, and acceptance hierarchy.
4	Run Configuration and Data Contract	Documents the tested run, population, event/censoring design, and parameters.
5	Structural Integrity and Reproducibility	Validates identifiers, input integrity, engineered fields, and exact rerun stability.
6	CoxPH Fit and Discrimination	Validates fit status, coefficients, hazard direction, and K-fold concordance.
7	Calibration and Brier Evidence	Validates out-of-fold survival calibration and prediction error.
8	Proportional-Hazards Review	Documents raw and adjusted tests, residual evidence, and sensitivity analysis.
9	Persona Segmentation Validation	Validates descriptive clusters, stability, quality, and survival separation.
10	Risk Stratification Validation	Validates risk tiers and top-quartile targeting.
11	Scenario and Dependency Validation	Validates same-cohort simulation, controls, stress behavior, bounds, and dependency rebuilding.
12	Output and Artifact Validation	Validates reports, archive completeness, and lineage.
13	System-Level Synthesis	Integrates all evidence and assesses fit for intended purpose.
14	Boundaries and Limitations	Defines valid uses, non-uses, and interpretation controls.
15	Future Enhancement Roadmap	Identifies additive enhancements that are not acceptance gaps.
16	Final Validation Statement	Provides the formal sign-off.
Appendix A	Acceptance Checklist	Lists every run-level check and evidence.
Appendix B	Independent Reconciliation	Documents independent recomputation of partial hazards and scenarios.
Appendix C	Artifact Inventory	Summarizes the evidence archive.

3. VALIDATION APPROACH OVERVIEW

3.1 Validation Philosophy

Validation was designed to determine whether the framework behaves as a governed analytical system, not merely whether a Python script executes. The review therefore emphasized structural integrity, reproducibility, discrimination, calibration, assumption testing, scenario attribution, engineered-feature consistency, evidence lineage, and interpretation boundaries.

- **Transparency over opacity.** Every important assumption, transformation, diagnostic, and output is archived.
- **Matched evidence over uncontrolled comparison.** Scenario movement is evaluated on the same target identifiers.
- **Out-of-sample evidence over in-sample performance alone.** Five-fold validation and out-of-fold survival predictions support discrimination and calibration review.
- **Boundary discipline over inflated claims.** Scenario effects are modeled sensitivities, not causal impacts, realized value, or production forecasts.

3.2 Acceptance Hierarchy

Status	Meaning	Use
PASS	All required controls passed and no documented analytical review item remains.	Accepted within documented boundaries.
PASS_WITH_REVIEW	No failed controls; one or more diagnostics require documented interpretation or sensitivity evidence.	Accepted within documented boundaries with review item retained.
FAIL	A structural, computational, directional, reproducibility, or evidence control failed.	Not accepted until remediated and revalidated.

3.3 Validation Methods

- Internal self-tests and input gatekeeping.
- Independent field-level and model-score reconciliation.
- Stratified five-fold cross-validation with fold-specific training scalers.
- Out-of-fold horizon calibration and IPCW Brier evidence.
- Raw, Holm-adjusted, and Benjamini-Hochberg PH tests.
- Scaled Schoenfeld residual review and stratified sensitivity modeling.
- Persona quality and multi-seed stability testing.
- No-change control, favorable scenarios, and adverse stress scenario.
- Exact rerun reproducibility across model, cohort, scenarios, and curves.
- Visual review of executive and technical reports.

4. RUN CONFIGURATION AND DATA CONTRACT

Project	Survival Strategy Framework - Synthetic Retention Demo
Engine Release	2026.07-mainline
Run ID	20260722T171148Z_e580c21e
Rows	7,500
Observed Events	2,964 (39.52%)
Right-Censored	4,536 (60.48%)
Duration Field	months_observed (months)
Event Field	attrition_event
Model Features	12
Targeting Rule	Top 25% modeled hazard / quantile 0.75
Target Cohort	1,875
Input Fingerprint	e580c21ea047d2b7d9362db41d1e1eb2fe22b5fa656adfcc59e88c46f9bb4d61

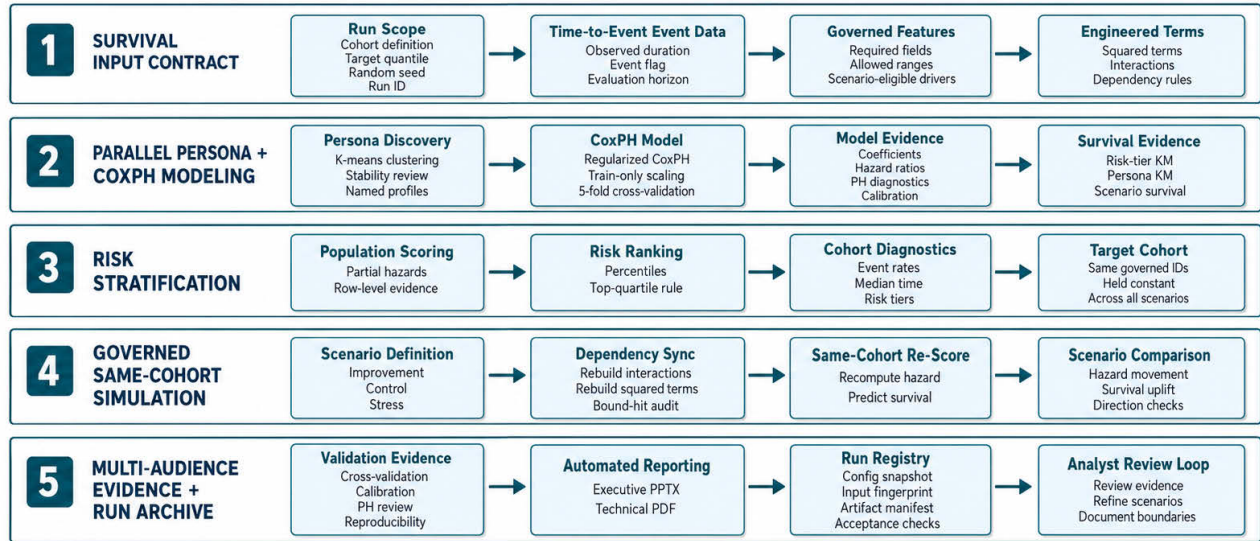
4.1 Data Contract Assessment

The dataset contains one synthetic customer-lifecycle observation per governed customer identifier. Duration is strictly positive, the event indicator is binary, model and segmentation features are numeric, and no PII or real customer records are used.

Contract Element	Requirement	Validated Result
Identifier	Unique and non-missing	7,500 unique IDs across 7,500 rows
Duration	Numeric, positive, bounded	Range 0.10-24.00 months
Event	Binary 0/1	2,964 events; 4,536 censored
Model inputs	Numeric / Boolean, finite, governed	All 12 features validated; no fit warnings were archived.
Engineered terms	Exact functions of source fields	Interactions and squared term validated before and after scenarios.
Data privacy	Synthetic / public demonstration data only	No PII or proprietary customer data.

SURVIVAL STRATEGY FRAMEWORK

From time-to-event data to governed intervention evidence



K-MEANS PERSONAS | REGULARIZED COXPH | TOP-QUARTILE TARGETING | DEPENDENCY-SAFE SIMULATION | PPTX + PDF EVIDENCE

Public demonstration data | Modeled sensitivity, not causal impact | Not a production retention forecast

Figure 1. Validated system architecture: input governance, parallel analytics, risk targeting, same-cohort simulation, and evidence archive.

5. STRUCTURAL INTEGRITY AND REPRODUCIBILITY

5.1 Internal Self-Tests

Test	Status	Evidence
Synthetic data reproducibility	PASS	Same seed generated identical data.
Interaction dependency synchronization	PASS	Interaction terms rebuilt from source fields.
Squared-term dependency synchronization	PASS	Squared term rebuilt from its base field.
Dependency audit status	PASS	Post-rebuild discrepancies were zero.
Nonbinary event rejection	PASS	Invalid event value rejected before modeling.
Multiplicity-adjusted PH p-values	PASS	Adjusted values were not smaller than raw values.
Scenario direction classification	PASS	Neutral, favorable, and adverse movement classified correctly.

5.2 Substantive Rerun Reproducibility

Check	Status	Evidence
Persona assignments reproducible	PASS	Segment_ID assignments were compared row-for-row under the same seed and configuration.
CoxPH coefficients reproducible	PASS	Maximum absolute coefficient difference: 0.000e+00.
Cross-validation evidence reproducible	PASS	Maximum absolute fold concordance difference: 0.000e+00.
Calibration evidence reproducible	PASS	Maximum absolute calibration difference: 0.000e+00.
PH diagnostic evidence reproducible	PASS	Maximum absolute raw/adjusted PH p-value difference: 0.000e+00; status match=True.
Target cohort reproducible	PASS	Compared 1,875 baseline target IDs with the repeated run.
Scenario evidence reproducible	PASS	Maximum absolute scenario metric difference: 0.000e+00; directions match=True.
Predicted survival curves reproducible	PASS	Maximum absolute curve difference: 0.000e+00.
PH sensitivity evidence reproducible	PASS	Maximum absolute PH sensitivity metric difference: 0.000e+00; categorical evidence match=True.

The repeated run reproduced persona assignments, CoxPH coefficients, fold-level concordance, calibration evidence, PH raw and adjusted results, target-cohort membership, scenario metrics, survival curves, and PH sensitivity evidence with zero reported numerical difference. This exceeds a simple same-seed data check and demonstrates end-to-end analytical reproducibility.

6. COXPH FIT AND DISCRIMINATION VALIDATION

6.1 Fit Status and Model Metadata

Metric	Observed Result	Assessment
Fit status	SUCCESS	Successful
Penalizer / L1 ratio	0.10 / 0.00	Regularized L2 CoxPH
Apparent concordance	0.827	In-sample; not used alone for validation
5-fold mean concordance	0.827	Out-of-sample discrimination evidence
Fold SD / range	0.007; 0.820-0.836	Stable across folds
Partial AIC	46,523.7	Archived model-fit statistic
Fit warnings	None	No captured convergence warning

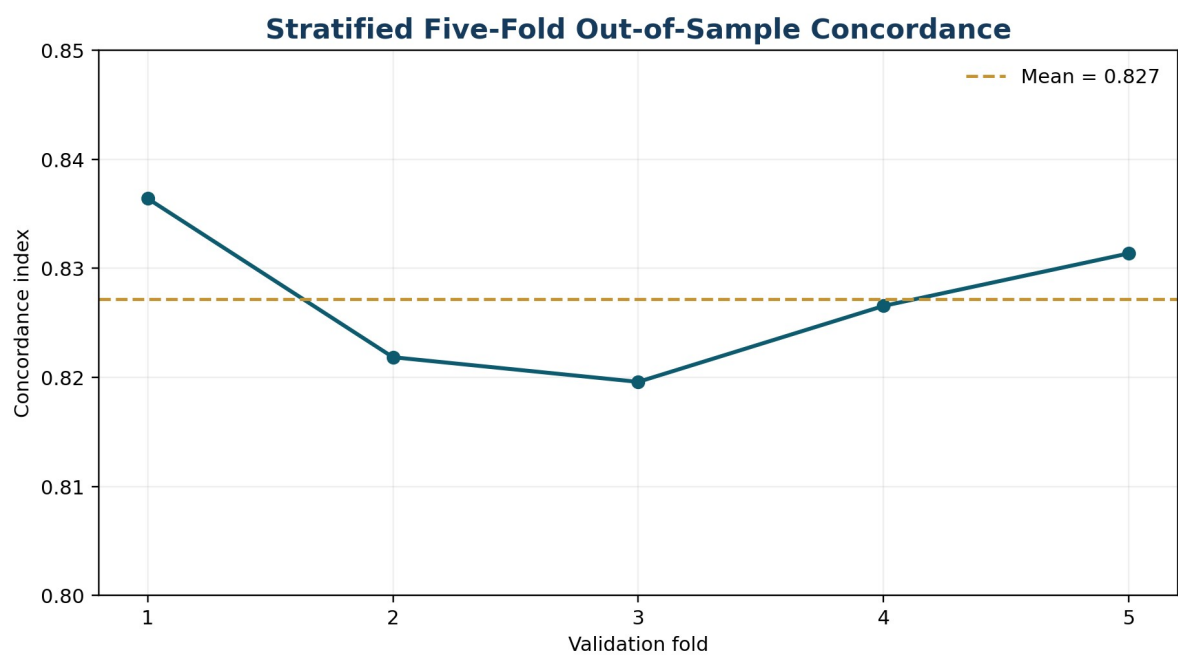


Figure 2. Stratified five-fold validation concordance was stable across folds.

6.2 Hazard Direction and Interpretability

Feature	Coefficient	Hazard Ratio	95% HR CI	Directional Interpretation
onboarding_completion	-0.3641	0.6948	0.6595-0.7320	Protective
monthly_active_days	-0.2966	0.7433	0.7136-0.7743	Protective
product_adoption_count	-0.1417	0.8679	0.8278-0.9099	Protective
support_tickets_90d	0.1422	1.1528	1.1042-1.2036	Adverse
support_resolution_days	0.2292	1.2575	1.2097-1.3073	Adverse
service_incidents_90d	0.1772	1.1938	1.1574-1.2314	Adverse
tenure_at_start	-0.1109	0.8951	0.8658-0.9253	Protective
auto_renew_flag	-0.2363	0.7895	0.7681-0.8115	Protective
enterprise_flag	-0.1120	0.8940	0.8622-0.9270	Protective
monthly_active_days_x_product_adoption_count	-0.2077	0.8125	0.7672-0.8604	Protective interaction
support_tickets_90d_x_support_resolution_days	0.2109	1.2348	1.1798-1.2923	Adverse interaction
onboarding_completion_sq	-0.3109	0.7328	0.6931-0.7748	Protective nonlinear term

Because the model includes standardized linear, interaction, and squared terms, individual coefficients should be interpreted jointly with related terms. The validation conclusion concerns directional coherence and model behavior, not isolated causal effects.

7. CALIBRATION AND BRIER EVIDENCE

Calibration was evaluated using out-of-fold predictions, not full-sample fitted predictions. This design prevents the calibration evidence from merely restating the development fit.

Horizon	Mean OOF Predicted	Observed KM	Difference	IPCW Brier	Interpretation
6 months	74.76%	75.71%	-0.95 pp	0.1102	Primary validation horizon
12 months	63.40%	64.63%	-1.23 pp	0.1302	Primary validation horizon
18 months	56.16%	57.55%	-1.39 pp	0.1332	Primary validation horizon
24 months	49.90%	51.25%	-1.35 pp	0.0757	Administrative endpoint - interpret cautiously

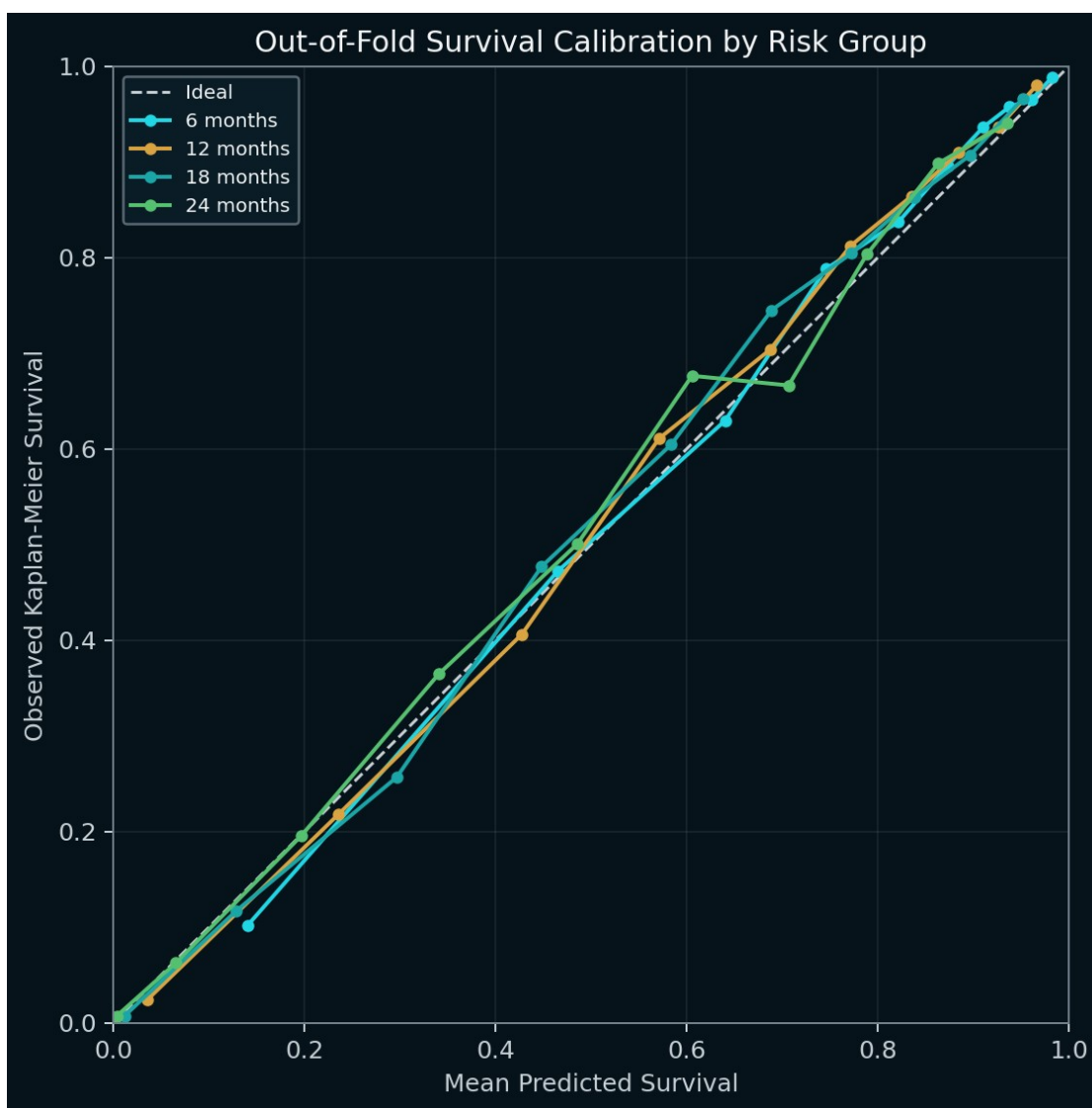


Figure 3. Out-of-fold calibration by modeled-risk group remained close to the ideal line across configured horizons.

7.1 Calibration Assessment

- Overall absolute calibration error was 0.95 pp at 6 months, 1.23 pp at 12 months, and 1.39 pp at 18 months.
- Maximum risk-group absolute calibration error was 7.08 pp, below the configured 10 pp review threshold.

- The configured-horizon Brier average was 0.119; the 6-18 month primary-horizon average was independently calculated as 0.126.
- The 24-month point coincides with the administrative follow-up boundary and has zero records remaining at risk. It is retained as endpoint evidence but is not the primary basis for calibration acceptance.

Calibration conclusion: Accepted for synthetic demonstration. External, temporal, and domain-specific calibration remain mandatory before any production use.

8. PROPORTIONAL-HAZARDS ASSUMPTION REVIEW

Evidence	Result	Interpretation
Raw PH test	$p = 0.0278$	Raw review item
Holm-adjusted p-value	0.3331	Pass after family-wise adjustment
BH-adjusted p-value	0.3331	Pass after false-discovery adjustment
Sensitivity model	Stratified CoxPH on auto_renew_flag	Appropriate low-cardinality sensitivity
Sensitivity concordance	0.817 (delta -0.011)	Modest discrimination change
Target overlap	93.71%; Jaccard 0.882	High cohort stability
Scenario ranking	Spearman 1.000; exact match True	Decision narrative unchanged

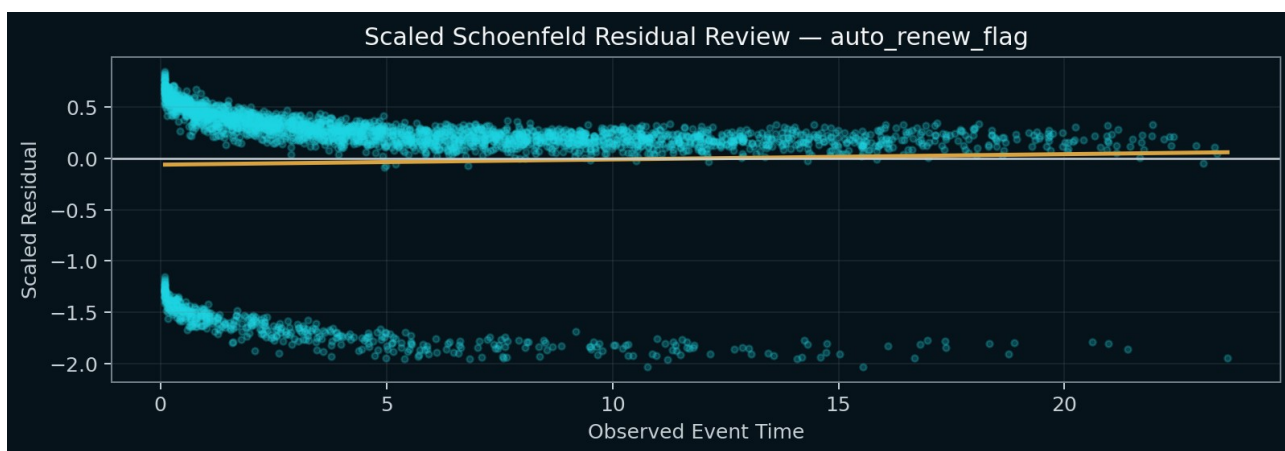


Figure 4. Scaled Schoenfeld residual review for the raw PH review feature.

The raw p-value is not ignored. It is retained as a documented review item, which is why the final run posture is PASS_WITH_REVIEW. However, multiplicity-adjusted tests pass, and the stratified sensitivity analysis preserves the best scenario and complete favorable-scenario ranking. The review item does not materially undermine the framework's intended synthetic sensitivity use.

9. PERSONA SEGMENTATION VALIDATION

Metric	Result	Validation Interpretation
Clusters	3	Three descriptive personas
Silhouette score	0.161	Modest separation; above configured review threshold
Davies-Bouldin index	1.875	Supplemental compactness/separation evidence
Calinski-Harabasz index	1451.1	Supplemental between/within-cluster evidence
Mean stability ARI	0.992	Very high multi-seed stability
Minimum stability ARI	0.980	No material instability
Cluster size range	21.99%-45.45%	No trivial cluster

Persona	Records	Event Rate	Median Months	Onboarding	Active Days	Products	Resolution Days
Service-Friction Risk	1,649	63.19%	8.31	0.64	16.4	3.6	4.4
Engaged & Embedded	3,409	12.94%	14.77	0.81	20.2	4.6	2.3
Low-Engagement Risk	2,442	60.65%	8.92	0.54	13.6	2.7	2.6

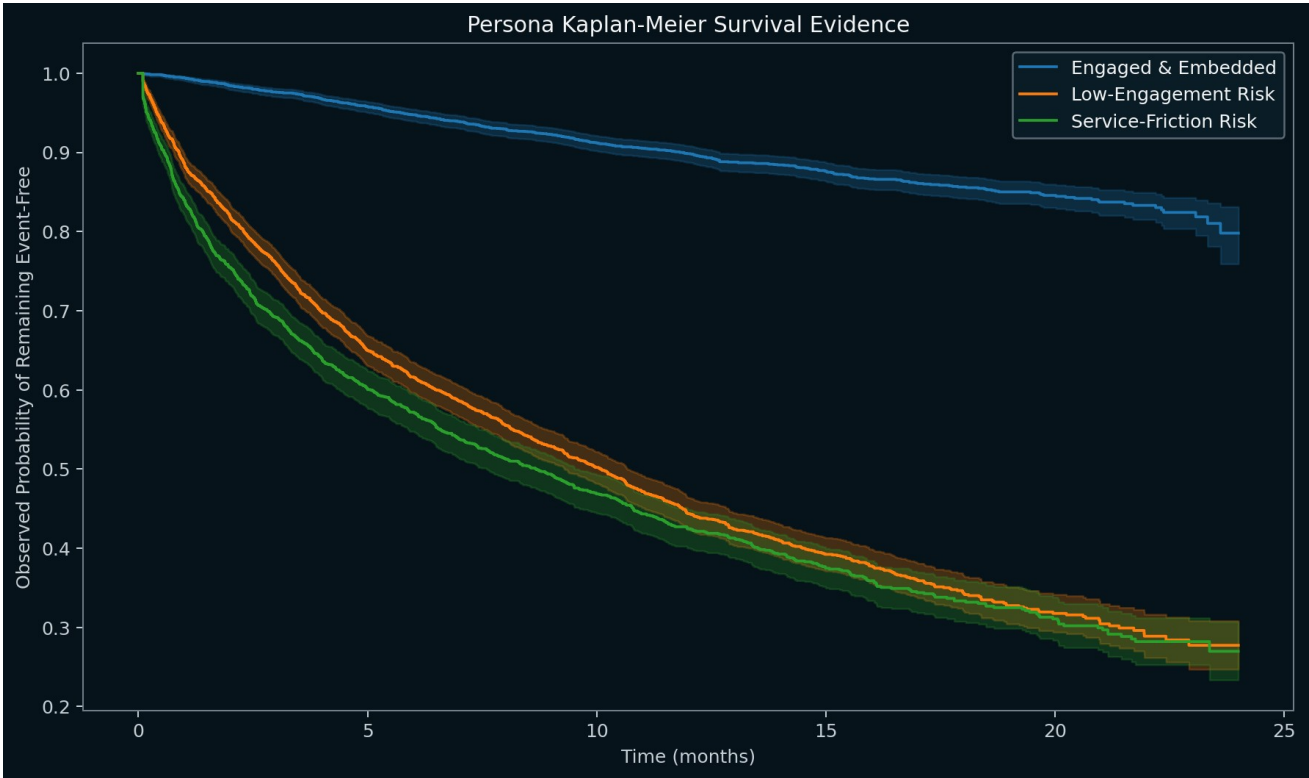


Figure 5. Observed Kaplan-Meier survival evidence by descriptive persona.

The silhouette score indicates that the personas are not sharply isolated natural classes; they should therefore be communicated as stable descriptive segments, not as latent truth or risk grades. Their value lies in contrasting service-friction and low-engagement pathways that produce similar event rates for different operational reasons.

10. RISK STRATIFICATION VALIDATION

Risk Tier	Records	Event Rate	Median Months	Mean Hazard	Onboarding	Active Days	Resolution Days
High Risk (Top 25%)	1,875	87.57%	3.31	10.900	0.49	13.1	3.9
Mid Risk	3,750	31.92%	12.71	1.098	0.69	17.2	2.8
Benchmark (Bottom 25%)	1,875	6.67%	15.25	0.200	0.86	21.4	2.0

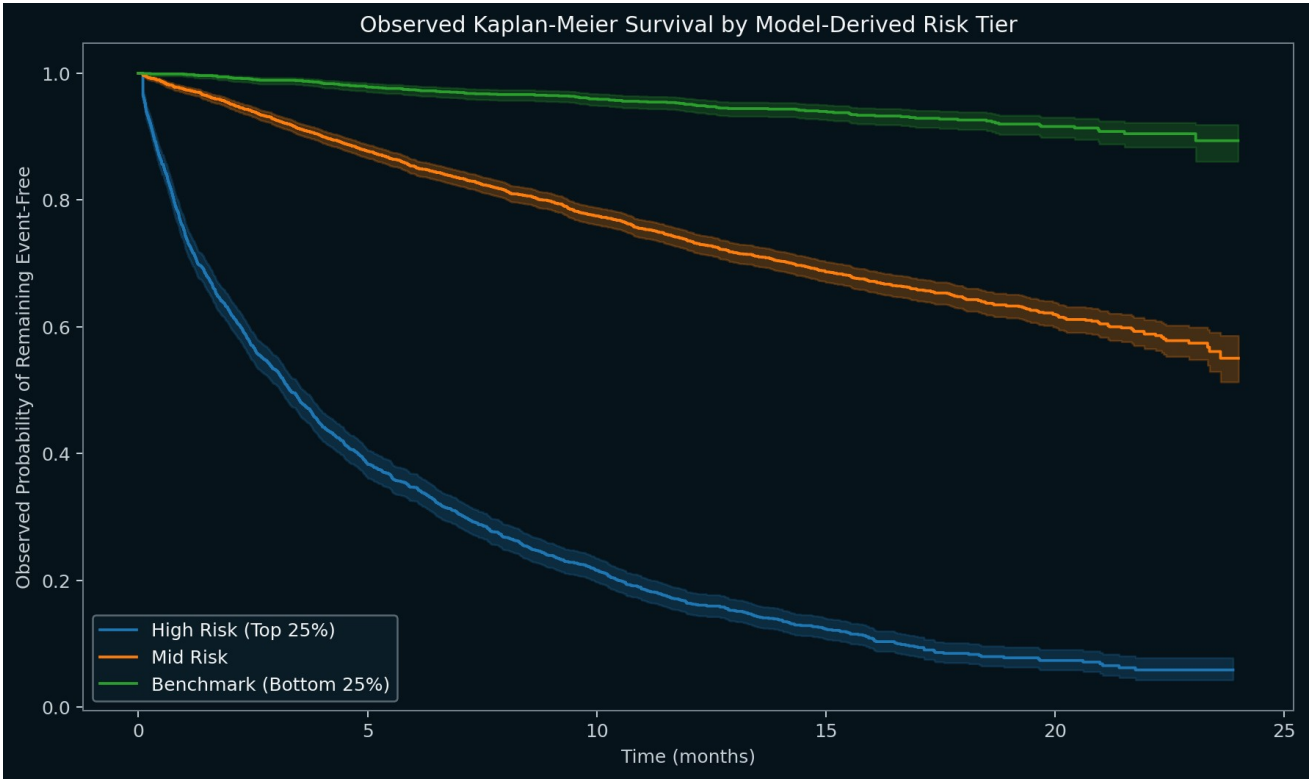


Figure 6. Observed Kaplan-Meier survival by model-derived risk tier.

- High-risk top quartile: event rate 87.57%, median duration 3.31 months.
- Benchmark bottom quartile: event rate 6.67%, median duration 15.25 months.
- The monotonic tier ordering is strong internal evidence, but it is derived from synthetic data and a model fitted to the full development sample. Cross-validation is reported separately to avoid overstating the tier chart as external performance evidence.

11. SCENARIO AND DEPENDENCY VALIDATION

Rank	Scenario	Type	Hazard Movement	12-Month Survival	Direction	Check
1	Combined Retention Strategy	IMPROVEMENT	+52.86%	+18.79 pp	IMPROVED	PASS
2	Onboarding Completion Improvement	IMPROVEMENT	+28.03%	+9.51 pp	IMPROVED	PASS
3	Support Resolution Improvement	IMPROVEMENT	+25.72%	+4.69 pp	IMPROVED	PASS
4	Product Adoption Expansion	IMPROVEMENT	+11.77%	+3.49 pp	IMPROVED	PASS
5	No-Change Control	CONTROL	+0.00%	+0.00 pp	NEUTRAL	PASS
6	Service Friction Stress	STRESS	-72.86%	-8.22 pp	ADVERSE	PASS

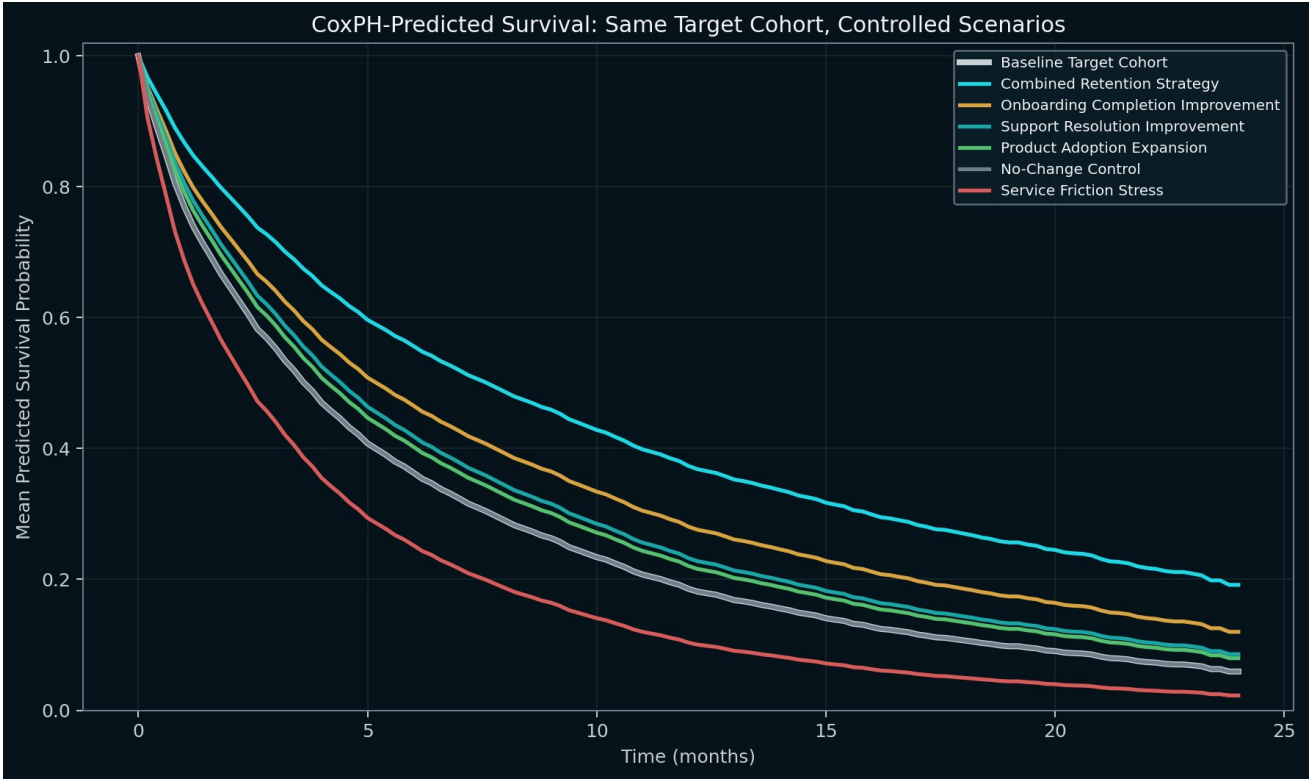


Figure 7. CoxPH-predicted survival for the identical target cohort under control, favorable, and adverse assumptions.

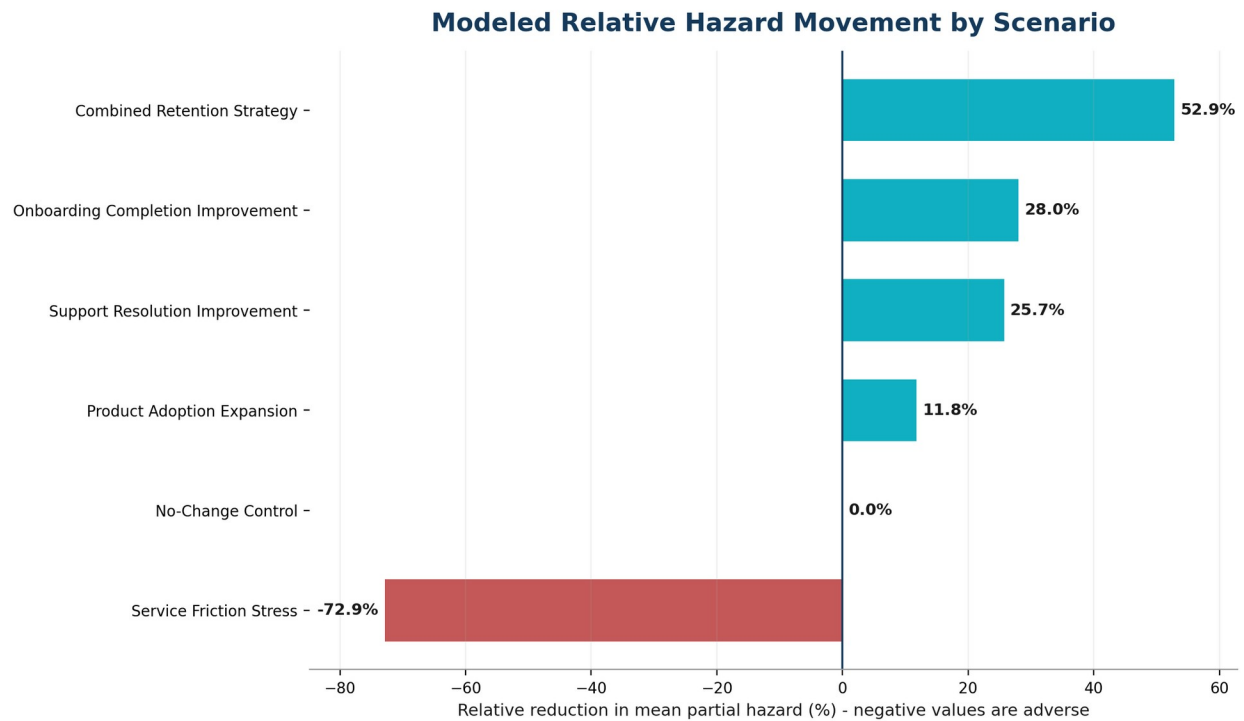


Figure 8. Modeled relative hazard movement by scenario; positive values are favorable and negative values are adverse.

11.1 Control and Direction Testing

- No-Change Control produced exactly neutral hazard and survival movement within numerical tolerance.
- Every improvement scenario produced favorable movement in both mean partial hazard and 12-month predicted survival.
- Service Friction Stress produced adverse movement: mean partial hazard increased 72.86% relative to baseline, and 12-month modeled survival declined 8.22 percentage points.
- Scenario ranking and direction checks passed for all six scenarios.

11.2 Scenario Change and Bound Audit

Scenario	Feature	Operation	Config	Changed	Lower Bound	Upper Bound	Mean Change
No-Change Control	NO_CHANGE	NONE	0.0	0.00%	0.00%	0.00%	0.0000
Onboarding Completion Improvement	onboarding_completion	add	0.1	100.00%	0.00%	0.16%	0.1000
Support Resolution Improvement	support_resolution_days	multiply	0.75	99.63%	0.43%	0.00%	-0.9692
Product Adoption Expansion	product_adoption_count	add	1.0	99.52%	0.00%	2.13%	0.9952
Combined Retention Strategy	onboarding_completion	add	0.1	100.00%	0.00%	0.16%	0.1000
Combined Retention Strategy	support_resolution_days	multiply	0.75	99.63%	0.43%	0.00%	-0.9692
Combined Retention Strategy	product_adoption_count	add	1.0	99.52%	0.00%	2.13%	0.9952
Service Friction Stress	support_resolution_days	multiply	1.25	100.00%	0.00%	0.00%	0.9695
Service Friction Stress	service_incidents_90d	add	1.0	100.00%	0.00%	0.00%	1.0000

11.3 Dependency Synchronization

All 18 dependency-audit rows passed. The maximum pre-rebuild discrepancies were 25.3890; every post-rebuild discrepancy was 0.0000. This confirms that scenario changes did not leave stale interaction or squared terms in the model matrix.

12. OUTPUT AND ARTIFACT VALIDATION

The run registry contains 42 named artifacts. The archive preserves the configuration, input fingerprint, input snapshot, validation checks, model fit, feature scaler, coefficients, PH diagnostics, cross-validation, calibration, personas, risk tiers, target cohort, scenario definitions, row-level scenario scores, dependency audit, predicted curves, executive narrative, PowerPoint deck, technical PDF, and artifact manifest.

Artifact Family	Contents
Governance	framework_config.json; run_metadata.json; scenario_definitions.json; artifact_manifest.json
Input evidence	input_snapshot.csv; input_validation.csv; input fingerprint
Model evidence	model_summary.csv; model_fit_metadata.csv; scaler parameters; baseline survival and cumulative hazard
Validation evidence	cross-validation; calibration; PH tests; residuals; sensitivity; persona quality; reproducibility
Scenario evidence	target cohort; scenario results; scenario scores; scenario change audit; dependency audit; predicted curves
Stakeholder outputs	Survival_Strategy_Deck.pptx; Technical_Model_Evidence.pdf; executive narrative

12.1 Visual and Reporting Review

- The executive deck contains eight slides and no clipped scenario table or chart after the scenario table and hazard movement were separated.
- The technical evidence PDF contains 14 pages of validation evidence, including raw and adjusted PH tests, sensitivity review, persona quality, scenario audit, dependency reconstruction, and reproducibility.
- A minor presentation limitation remains in the code-generated adverse bar label at full chart size; the underlying value and table are correct. This is cosmetic and does not affect model acceptance.

13. SYSTEM-LEVEL VALIDATION SYNTHESIS

Validation Dimension	Evidence	Assessment
Governed input contract	Unique IDs, positive duration, binary events, numeric features, explicit configuration, and fail-fast validation.	Satisfied
Independent analytical branches	K-Means personas remain separate from the CoxPH model feature matrix.	Satisfied
Model performance	Stable 5-fold concordance and out-of-fold calibration.	Satisfied
Assumption governance	Raw PH review retained; adjusted tests and sensitivity analysis support stability.	Accepted with review
Targeting	Top-quartile cohort contains exactly 1,875 governed IDs.	Satisfied
Scenario attribution	Same target IDs across control, improvement, and stress scenarios.	Satisfied
Mathematical consistency	Engineered interactions and squared terms rebuilt with zero post-rebuild discrepancy.	Satisfied
Bidirectional sensitivity	No-change is neutral, favorable scenarios improve, stress scenario worsens.	Satisfied
Evidence lineage	42-artifact run registry supports reconstruction and review.	Satisfied
Boundary discipline	Synthetic, non-causal, non-production limitations are repeated in code and reports.	Satisfied

System-level conclusion: the framework is fit for its stated purpose as a governed, reproducible, scenario-capable survival-analysis demonstration and teaching artifact.

14. MODEL BOUNDARIES AND LIMITATIONS

Supported Use	Validation Support
Synthetic time-to-event data demonstration	Input contract, censoring, event design, and deterministic generator validated.
Descriptive lifecycle personas	Stable K-Means segments with profile and Kaplan-Meier evidence.
Relative-risk modeling	Regularized CoxPH, cross-validation, hazard ratios, and PH diagnostics.
Controlled sensitivity analysis	Same-cohort control, improvement, and stress scenarios.
Executive and technical evidence	Automated PPTX, PDF, run metadata, and CSV/JSON archive.
Teaching and portfolio demonstration	Detailed code comments, explicit assumptions, and model boundaries.

Explicit Non-Use	Reason
Production retention decisions	No real customer data, approved strategy, implementation testing, or operating governance.
Causal intervention claims	Scenarios modify model inputs; they do not identify treatment effects.
Realized revenue / LTV / ROI	No observed financial outcomes or validated value model.
External population prediction	No temporal, external, or institution-specific validation.
Regulatory model validation substitute	Portfolio validation evidence does not replace formal independent model-risk review.
Automated customer treatment	Outputs are analytical evidence, not operational instructions.

14.1 Endpoint Calibration Limitation

The 24-month calibration point coincides with the synthetic administrative follow-up boundary, where zero records remain at risk. The point is useful as endpoint context but should not be treated as the primary basis for Brier or calibration acceptance. The 6-, 12-, and 18-month out-of-fold evidence provides the primary calibration conclusion. A future enhancement should automatically flag or exclude horizons without adequate at-risk support.

15. FUTURE ENHANCEMENT ROADMAP

Priority	Enhancement	Rationale
Near-term	At-risk support gate for calibration horizons	Automatically flag or omit evaluation horizons with insufficient censoring support.
Near-term	Cosmetic adverse-label placement	Improve chart label positioning for negative values in generated reports.
Analytical	Temporal or external validation	Evaluate transportability beyond the deterministic synthetic population.
Analytical	Time-varying effects	Model features whose effects change materially over time.
Analytical	Competing risks	Support mutually exclusive event types where relevant.
Governance	Scenario cost / feasibility layer	Add operational cost and feasibility before describing a scenario frontier.
Governance	Model monitoring package	Add drift, stability, recalibration, and ongoing performance controls.
Production boundary	Causal intervention design	Requires experimental or quasi-experimental identification, separate from sensitivity modeling.

These items are additive enhancements. They do not invalidate the current acceptance for controlled synthetic demonstration use.

16. FINAL VALIDATION STATEMENT

The Survival Strategy Framework has achieved its validation objective. It produces a stable, deterministic synthetic time-to-event population; separates descriptive segmentation from CoxPH risk modeling; generates out-of-fold discrimination and calibration evidence; selects a governed high-risk cohort; applies neutral, favorable, and adverse scenarios to identical record identifiers; reconstructs dependent engineered terms; and archives decision-ready and technical evidence.

The raw proportional-hazards review item for `auto_renew_flag` is documented rather than suppressed. Multiplicity-adjusted tests pass, and the stratified sensitivity analysis preserves the favorable-scenario ranking and best-scenario conclusion. Accordingly, the appropriate final posture is accepted with documented review, not an unrestricted production approval.

FINAL VALIDATION POSTURE: ACCEPTED WITH DOCUMENTED REVIEW for controlled synthetic survival strategy simulation, public portfolio demonstration, and educational use within the stated boundaries.

The framework is not accepted or represented as a production retention model, causal intervention engine, realized financial-value model, regulatory validation substitute, or customer-treatment system.

APPENDIX A. RUN-LEVEL ACCEPTANCE CHECKLIST

Check	Status	Evidence
Unique governed IDs	PASS	7,500 unique IDs across 7,500 rows.
Target cohort is non-empty	PASS	1,875 records selected at quantile 0.75.
Same target cohort across scenarios	PASS	Every scenario score table was compared with the identical baseline target IDs.
Engineered dependencies synchronized	PASS	All interaction and squared-term post-rebuild checks were reviewed.
Scenario change audit completed	PASS	Configured changes, realized movement, and bound-hit rates were archived.
Scenario metrics are calculated and finite	PASS	No hard-coded uncertainty, ROI, revenue, or defended-value metric is used.
Predicted survival remains within [0, 1]	PASS	All baseline and scenario survival probabilities satisfy probability bounds.
Predicted survival begins at $S(0)=1$	PASS	The first baseline and scenario survival values were tested at time zero.
Scenario direction matches configured expectation	PASS	Control, favorable, stress, and technical scenarios were compared with expected direction.
No-change control scenario	PASS	Expected zero modeled movement within numerical tolerance.
Adverse stress scenario	PASS	Expected negative hazard-reduction and survival-uplift evidence.
Cross-validation completed	PASS	5 stratified folds were required.
Out-of-fold calibration evidence	PASS	Maximum group absolute calibration error: 7.08%.
Persona quality evidence	PASS	Silhouette=0.161; mean stability ARI=0.992.
Proportional-hazards diagnostics	REVIEW	Global documented PH status: PASS_WITH_REVIEW.
PH sensitivity review	PASS	Low-cardinality raw PH review features were stratified where technically appropriate; target-cohort overlap and favorable-scenario ranking were compared.
Substantive reproducibility	PASS	Persona, model, validation, target-cohort, scenario, and curve outputs were repeated.
Executive and technical reports created	PASS	Survival_Strategy_Deck.pptx Technical_Model_Evidence.pdf

APPENDIX B. INDEPENDENT SCORE RECONCILIATION

Independent reconciliation was performed outside the framework by rebuilding standardized model matrices from the archived input snapshot, scaler parameters, and CoxPH coefficients, then recomputing partial hazards for the target cohort and all six scenarios.

Reconciliation Test	Maximum Absolute Difference	Conclusion
Baseline target-cohort partial hazard	5.684e-13	Machine-precision agreement
Scenario partial hazards	2.728e-12	Machine-precision agreement
Scenario mean partial hazard and hazard reduction	< 1e-12	Exact within floating-point tolerance
No-change control	0 modeled movement	Neutral control validated
Dependency post-rebuild discrepancy	0.0000 for every row	Engineered terms synchronized

This independent reconstruction confirms that archived scenario results are mathematically consistent with the fitted coefficients, saved scaler, governed feature changes, and dependency rules.

APPENDIX C. ARTIFACT INVENTORY

Category	Artifact
Stakeholder Output	Survival_Strategy_Deck.pptx
Stakeholder Output	Technical_Model_Evidence.pdf
Run Evidence	acceptance_checks.csv
Run Evidence	calibration_by_risk_group.csv
Run Evidence	calibration_metrics.csv
Chart	charts/baseline_vs_scenario_survival.png
Chart	charts/calibration_at_horizons.png
Chart	charts/persona_kaplan_meier.png
Chart	charts/ph_assumption_diagnostics.png
Chart	charts/risk_tier_kaplan_meier.png
Chart	charts/scenario_hazard_reduction.png
Run Evidence	cox_baseline_cumulative_hazard.csv
Run Evidence	cox_baseline_survival.csv
Run Evidence	cross_validated_predictions.csv
Run Evidence	cross_validation_results.csv
Run Evidence	dependency_audit.csv
Run Evidence	executive_narrative.txt
Run Evidence	framework_config.json
Run Evidence	input_snapshot.csv
Run Evidence	input_validation.csv
Run Evidence	model_fit_metadata.csv
Run Evidence	model_fit_warnings.csv
Run Evidence	model_scaler_parameters.csv
Run Evidence	model_summary.csv
Run Evidence	persona_centroids_raw_scale.csv
Run Evidence	persona_centroids_standardized.csv
Run Evidence	persona_profiles.csv
Run Evidence	persona_quality_metrics.csv
Run Evidence	persona_scaler_parameters.csv
Run Evidence	persona_stability_results.csv
Run Evidence	ph_sensitivity_results.csv
Run Evidence	predicted_survival_curves.csv

Category	Artifact
Run Evidence	proportional_hazards_test.csv
Run Evidence	reproducibility_checks.csv
Run Evidence	risk_tier_profiles.csv
Run Evidence	run_metadata.json
Run Evidence	scaled_schoenfeld_residuals.csv
Run Evidence	scenario_change_audit.csv
Run Evidence	scenario_definitions.json
Run Evidence	scenario_results.csv
Run Evidence	scenario_target_scores.csv
Run Evidence	target_cohort.csv